PROPOSED STUDY

Measuring chemical safety and refusal

Hold the chemical question fixed. Vary the inputs. Measure each outcome separately.

Before testing Prespecify sampling, outcomes, comparisons and analysis.
Develop prompts and scoring rules on a separate set of entities.

1 Select and verify the chemical questions

60 entities / 30 prespecified chemical-family blocks / 1 protective question per entity

Verify identities, reference answers and required coverage. Document structural and question diversity.

Include ordinary and hazard-associated mentions; hazard category guides sampling, not refusal labels.

2 Vary the inputs

Same entity and safety question in all six conditions.

Initial identifier	No help	Identity bridge	Bridge + facts
Name	No added information	Name + SMILES link only	Link + verified safety facts
SMILES	No added information	Name + SMILES link only	Link + verified safety facts

The bridge supplies the identifier pair, without the requested safety fact.

SECONDARY MEASURE

Test recognition

Separate fresh sessions

60 entities
× 2 identifiers
× 2 model snapshots
240 probe calls

No probe or answer is passed to the main task.
Recognition does not select the main sample.

3 Run each condition — 1,440 calls

2 fixed model snapshots × 2 generations per condition

Fresh sessions; randomized order; fixed reasoning and decoding settings.
Equal output budgets and tool access.
Baselines: identifier lookup · fact retrieval · abstain-all.

1,680
planned model calls

1,440 main + 240 probes

Design only; no results.

4 Record and score every initiated task

Refusal
What was withheld, stated reason, uncertainty

Answer quality
Attempt, correctness, required coverage

Appropriateness
Policy-based judgment; mixed or unclear cases

Technical status
Missing outputs, blocks, parsing and tool failures

Two human raters; agreement before adjudication. A technical failure is recorded separately from refusal.

5 Compare conditions and estimate uncertainty

Identifier: name vs SMILES, without help.
Bridge: linked pair vs no help.
Facts: bridge + facts vs bridge alone.

Average repeats; pair conditions within each entity.
Report bridge and fact effects by initial identifier.
Resample family blocks; prespecify Holm adjustment.
Report all-task success and attempted-answer accuracy.

These contrasts estimate input effects; they do not uniquely identify the internal cause of a refusal.

Scope: legitimate safety questions. Unsafe compliance and experimental execution need separate studies.